

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-accedad891d1ad93e.jsonl`  
- SHA-256: `8c006b8bddaed2373f47dd177ffcb66edc7119ba563cd027f1021ff3d6d108`  
- Source modified: `2026-06-16T03:19:16+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:17:48.905Z)

Web Searcher: Small-data / weak / timestamp-supervised TAL + active learning

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3-5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Your angle: ****Small-data / weak / timestamp-supervised TAL + active learning**** ? Dimension 3: maximizing quality per label with ~6716 videos ? point/timestamp supervision, active-learning video/segment selection, and pseudo-labeling, all directly applicable to the golden-set bottleneck.

Search query: `weakly supervised temporal action localization timestamp supervision SF-Net active learning few labeled videos pseudo-labeling self-training core-set 2023 2024`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-16T03:17:51.742Z)

I'll search for relevant research on small-data weak/timestamp-supervised TAL and active learning techniques.

3. user (2026-06-16T03:18:04.890Z)

Web search results for query: "timestamp supervision temporal action localization point-level annotation video segmentation 2023 2024 arXiv"

Links: [{"title": "Action-Agnostic Point-Level Supervision for Temporal Action Detection", "url": "https://arxiv.org/pdf/2412.21205"}, {"title": "Temporal Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/pdf/2103.06669"}, {"title": "[2310.05511] Proposal-based Temporal Action Localization with Point-level Supervision", "url": "https://arxiv.org/abs/2310.05511"}, {"title": "[2103.06669] Temporal Action Segmentation from Timestamp Supervision", "url": "https://arxiv.org/abs/2103.06669"}, {"title": "Point-Level Temporal Action"}]

Localization: Bridging Fully-supervised Proposals to Weakly-supervised Losses | Request PDF", "url": "https://www.researchgate.net/publication/347300936_Point-Level_Temporal_Action_Localization_Bridging_Fully-supervised_Proposals_to_Weakly-supervised_Losses"}, {"title": "OnlineTAS: An Online Baseline for Temporal Action Segmentation", "url": "https://arxiv.org/pdf/2411.01122"}, {"title": "[2212.11694] Timestamp-Supervised Action Segmentation from the Perspective of Clustering", "url": "https://arxiv.org/abs/2212.11694"}, {"title": "Point-Supervised Skeleton-Based Human Action Segmentation", "url": "https://arxiv.org/pdf/2603.06201"}, {"title": "SF-Net: Single-Frame Supervision for Temporal Action Localization", "url": "https://arxiv.org/pdf/2003.06845"}]

The search has returned several highly relevant results about temporal action localization, video segmentation, and point-level/timestamp supervision from arXiv. Here are the key findings:

Recent Research Highlights (2023-2024):

****Point-Level Supervised Temporal Action Localization:****

Point-level supervised temporal action localization (PTAL) aims at recognizing and localizing actions in untrimmed videos where only a single point (frame) within every action instance is annotated in training data. A novel proposal-based method localizes actions by generating and evaluating action proposals of flexible duration that involve more comprehensive temporal information. (arXiv 2310.05511, October 2023)

****Timestamp-Supervised Action Segmentation:****

Video action segmentation under timestamp supervision has recently received much attention due to lower annotation costs. (arXiv 2212.11694, April 2023)

****Foundational Work:****

The temporal action segmentation task uses timestamp supervision, which requires a comparable annotation effort to weakly supervised approaches, and yet provides a more supervisory signal. (arXiv 2103.06669, March 2021)

****Recent 2024 Work:****

SQL-Net: Semantic query learning for point-supervised temporal action localization and a stepwise multi-grained boundary detector for point-supervised temporal action localization in the European Conference on Computer Vision represent recent advances in this field.

These papers collectively demonstrate an active research area focusing on reducing annotation burden while maintaining model performance through point-level and timestamp-based supervision methods.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

4. user (2026-06-16T03:18:08.119Z)

Web search results for query: "weakly supervised temporal action localization timestamp supervision SF-Net active learning few labeled videos pseudo-labeling self-training core-set 2023 2024"

Links: [{"title": "Learning Reliable Dense Pseudo-Labels for Point-Level Weakly-Supervised Action Localization | Neural Processing Letters", "url": "https://link.springer.com/article/10.1007/s11063-024-11598-w"}, {"title": "Rethinking Pseudo-Label Guided Learning for Weakly Supervised Temporal Action Localization from the Perspective of Noise Correction", "url": "https://arxiv.org/pdf/2501.11124"}, {"title": "Action-Agnostic Point-Level Supervision for Temporal Action Detection", "url": "https://arxiv.org/pdf/2412.21205"}, {"title": "Weakly Supervised Temporal Action Localization with Segment-Level Labels", "url": "https://arxiv.org/pdf/2007.01598"}, {"title": "Weakly-Supervised Temporal Action Localization by ...", "url": "https://ojs.aaai.org/index.php/AAAI/article/view/28516/29006"}, {"title": "LPR: learning point-level temporal action localization through re-training | Multimedia Systems | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s00530-023-01128-4"}, {"title": "Weakly

supervised temporal action localization with actionness-guided false positive suppression - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/abs/pii/S0893608024002314"}, {"title": "Weakly Supervised Temporal Action Localization with Segment-Level Labels | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-030-88004-0_4"}]

Based on the web search results, I found relevant research on weakly supervised temporal action localization with the specific techniques and concepts you mentioned:

Key Findings:

****SF-Net and Timestamp Supervision:****

SF-Net focuses on single-frame supervision for temporal action localization, and point-level annotations refer to annotating only a single timestamp and action class for each action instance during training.

****Pseudo-Labeling and Dense Labels:****

Researchers have proposed both sparse pseudo-labels based methods and dense pseudo-labels based methods, with SF-Net mining sparse pseudo-labels through segment approaches, and current methods primarily focusing on mining sparse pseudo-labels and generating dense pseudo-labels.

****Self-Training Approaches:****

The STA branch is trained using a self-training approach that can generate action scores independent of video labels.

****Recent Work (2023-2024):****

The search results include several recent papers on point-level weakly-supervised temporal action localization:

- A method based on local representation enhancement and global temporal optimization comprising modules that enhance action feature representation and improve class activation sequence classification reliability (published April 2024)
- An approach using a feature embedding module based on transformer encoders to capture global context features and optimize pseudo-labels' quality through point-level annotations (published July 2023)

However, the search results do not contain specific information about active learning strategies or core-set selection methods in this domain that you may be looking for.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. assistant (2026-06-16T03:18:10.597Z)

Let me search for active learning and self-training specifically, plus distillation/teacher-student governance.

6. user (2026-06-16T03:18:19.101Z)

Web search results for query: "active learning temporal action localization video selection uncertainty diversity core-set annotation budget 2023 2024"

Links: [{"title": "Towards Active Learning for Action Spotting in Association Football Videos", "url": "https://arxiv.org/pdf/2304.04220"}, {"title": "Boundary-Centric Active Learning for Temporal Action Segmentation", "url": "https://arxiv.org/html/2604.15173v1"}, {"title": "Active learning method for temporal action localization in untrimmed videos", "url": "https://image-pubs.uspto.gov/dirsearch-public/print/downloadPdf/10726313"}, {"title": "Active Learning from Scene Embeddings for End-to-End Autonomous Driving", "url": "https://arxiv.org/pdf/2503.11062"}, {"title": "OmniViD: Omni-supervised active learning for video action detection", "url": "https://arxiv.org/pdf/2508.13983"}, {"title": "Two-Stage Active Learning for Efficient Temporal Action Segmentation Yuhao Su", "url": "https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/06348.pdf"}]

The web search has returned several relevant results about active learning for temporal action localization and segmentation. Here's a summary of the findings:

Key Research Areas

****Active Learning Approaches for Temporal Video Analysis:****

While active learning has been extensively analyzed on generic setups, only a few works apply those approaches to temporal video analysis. Brandla et al. proposed an active learning method for temporal activity localization (TAL) algorithms, based on uncertainty sampling.

****Selection Strategies:****

The research identifies several key strategies:

1. ****Uncertainty-Based Selection****: Active learning algorithms primarily focus on uncertainty and data diversity, with uncertainty-based methods prioritizing the selection of samples with the highest uncertainty in model predictions.
2. ****Diversity Maximization****: Sener et al. modeled the selection process as a core-set problem, sampling representative subsets of images by minimizing the L2 distance with the remaining samples in the dataset.
3. ****Boundary-Centric Approaches****: B-ACT combines uncertainty-based video selection with a boundary score that fuses local uncertainty, class ambiguity, and temporal prediction dynamics, querying only a few boundary frames per video while training with boundary-centered context.

****Recent Work (2024-2026):****

B-ACT demonstrates strong label efficiency and consistent gains over active learning baselines under extremely sparse budgets. The research shows that annotation budget efficiency is a critical focus area in recent temporal action analysis work.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-16T03:18:21.595Z)

Web search results for query: "few-shot temporal action detection data-efficient label-efficient TAL pseudo-label noise correction 2024 2025"

Links: [{"title": "Pseudo label refining for semi-supervised temporal action localization | PLOS One", "url": "https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0318418"}, {"title": "Pseudo label refining for semi-supervised temporal action localization - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11798530/"}, {"title": "Rethinking Pseudo-Label Guided Learning for Weakly Supervised Temporal Action Localization from the Perspective of Noise Correction", "url": "https://arxiv.org/pdf/2501.11124"}, {"title": "Rethinking Pseudo-Label Guided Learning for Weakly Supervised Temporal Action Localization from the Perspective of Noise Correction", "url": "https://arxiv.org/html/2501.11124v2/"}, {"title": "Pseudo label refining for semi-supervised temporal action localization", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11798530/"}, {"title": "Rethinking Pseudo-Label Guided Learning for Weakly Supervised Temporal Action Localization from the Perspective of Noise Correction | Proceedings of the AAAI Conference on Artificial Intelligence", "url": "https://ojs.aaai.org/index.php/AAAI/article/view/33094"}, {"title": "[Quick Review] Rethinking Pseudo-Label Guided Learning for Weakly Supervised Temporal Action Localization from the Perspective of Noise Correction", "url": "https://liner.com/review/rethinking-pseudolabel-guided-learning-for-weakly-supervised-temporal-action-localization"}]

I found several relevant research papers on temporal action detection with pseudo-label noise correction from 2024-2025:

Key Findings:

****1. Pseudo Label Refining (PLR) for Semi-Supervised TAL****

The PLR method uses a teacher-student framework with three key components: pseudo-label self-refinement with temporal region interesting pooling to improve boundary accuracy, a boundary synthesis module to further refine temporal intervals with multiple inference, and an adaptive weight learning strategy for progressively learning pseudo labels with different qualities. Using ActionFormer and BMN as detectors, this method achieved significant improvements on THUMOS14 and ActivityNet v1.3 datasets at label rates of 10% to 60%. This paper was published in February 2025.

****2. NoCo: Noise Correction Framework for Weakly Supervised TAL****

NoCo is a weakly-supervised temporal action localization method that incorporates a two-stage noisy label learning strategy to harness useful signals from noisy pseudo-labels. The method includes a Context-aware Label Augmentation module for rectifying inaccurate action boundaries, and a teacher-student-based online noise correction framework addressing short-action omissions and many-to-one problems through Missing Instance Compensation, Ambiguous Instance Correction, and High-quality Pseudo-label Mining. The method's performance rivals or exceeds some fully-supervised approaches like TAL-Net on THUMOS14. This AAAI 2025 paper was published in January 2025.

Both approaches address the critical challenge of noise in pseudo-labels for data-efficient temporal action localization, representing recent advances in the field.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-16T03:18:24.699Z)

I have strong coverage now. Let me do one more targeted search on the teacher-student distillation / knowledge distillation boundary for using a strong teacher at inference, since that maps to the Gemini governance question.

9. user (2026-06-16T03:18:34.476Z)

Web search results for query: "semi-supervised temporal action localization teacher-student mean teacher consistency self-training few labeled videos THUMOS ActivityNet 2023"

Links: [{"title":"Spatiotemporal Perturbation Based Dynamic Consistency for Semi-supervised Temporal Action Detection | SpringerLink","url":"https://link.springer.com/chapter/10.1007/978-3-030-98358-1_15"}, {"title":"Pseudo label refining for semi-supervised temporal action localization","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11798530/"}, {"title":"[PDF] Self-Supervised Learning for Semi-Supervised Temporal Action Proposal | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Self-Supervised-Learning-for-Semi-Supervised-Action-Wang-Zhang/b01db336a8ead672700a406f5c12d1edb9fb2697"}, {"title":"Learning Temporal Action Proposals With Fewer Labels","url":"https://arxiv.org/pdf/1910.01286"}, {"title":"Pseudo label refining for semi-supervised temporal action localization | PLOS One","url":"https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0318418"}, {"title":"Pseudo label refining for semi-supervised temporal action localization - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11798530/"}, {"title":"Self-Supervised Learning for Semi-Supervised Temporal Action Proposal | Request PDF","url":"https://www.researchgate.net/publication/355885702_Self-Supervised_Learning_for_Semi-Supervised_Temporal_Action_Proposal"}, {"title":"Stable Mean Teacher for Semi-supervised Video Action Detection","url":"https://arxiv.org/html/2412.07072"}, {"title":"Towards Adaptive Pseudo-label Learning for Semi-Supervised Temporal Action Localization","url":"https://arxiv.org/html/2407.07673v1"}]

The search returned relevant results on semi-supervised temporal action localization. Here's what I found:

Key Research Areas

Most existing semi-supervised learning methods for temporal action localization follow the

paradigm of self-training and consistency regularization, where self-training methods generate pseudo labels for retraining while consistency regularization applies data augmentation and requires consistent predictions.

Teacher-Student Framework Approaches

Recent methods propose pseudo-label refining (PLR) based on the teacher-student framework, with several key components. These include pseudo-label self-refinement using temporal region pooling to improve boundary accuracy, and a boundary synthesis module to refine temporal intervals with multiple inference.

One semi-supervised framework extends the Mean Teacher model with sequential perturbations for video understanding, achieving similar performance to fully-supervised approaches when trained with only 60% of labels.

Evaluation on Standard Benchmarks

Methods have been tested on THUMOS14 and ActivityNet v1.3 datasets, with results showing improvements of up to 3.5% mAP under different label rates (10%, 20%, 40%, and 60%).

Recent Work

Recent research from 2023 includes work on sparse action detectors, and Mean Teacher based approaches have demonstrated superior performance among consistency-based methods, extending student-teacher learning to video domain for spatio-temporal detection tasks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. assistant (2026-06-16T03:18:37.505Z)

I now have comprehensive coverage across all sub-areas of Dimension 3: timestamp/point-level supervision, active learning with core-set/boundary selection, semi-supervised self-training/teacher-student, and pseudo-label noise correction. Let me select the highest-leverage results.

11. user (2026-06-16T03:19:05.680Z)

Structured output provided successfully

12. assistant (2026-06-16T03:19:16.144Z)

I've completed the research and returned the structured output. Here is a summary of the six highest-leverage results for Dimension 3 (small-data / weak / timestamp-supervised TAL + active learning):

****Timestamp / point-level supervision (cheapest-label wins):****

- ****Timestamp Supervision for Action Segmentation**** (Li et al., CVPR 2021) ? the canonical framing that matches the project's actual bug (a segmentation/boundary failure, not a recall failure); one click per rally recovers near-full-supervision boundaries.
- ****SF-Net**** (ECCV 2020) ? the named single-frame baseline; its pseudo-background mining is the conceptual tool for teaching the scorer that "shuttle moving during dead-time" is not rally.

****Active learning (which video/segment to label next under a tiny budget):****

- ****Two-Stage Active Learning for TAS**** (Su et al., ECCV 2024) ? select informative videos, then informative frames within them; maps cleanly onto leave-one-video-out and the Roora 6?16 growth.
- ****B-ACT: Boundary-Centric Active Learning**** (2026) ? fuses uncertainty video selection with a boundary score, querying only a few boundary frames; uniquely addresses both the over-merge problem and the label budget at once (newest, treat as a research bet).

****Pseudo-label / self-training with noise handling (the Gemini-governance-relevant half):****

- ****NoCo**** (AAAI 2025) ? frames automatically/teacher-derived labels as noise to be corrected;

reinforces the constraint that a paid-LLM teacher stays inference/eval-only, never owned-weight training data.

- ****PLR**** (PLOS One, Feb 2025) ? boundary-aware teacher-student self-training template to bootstrap the scorer from heuristic candidates using an in-domain (license-clean) teacher.

All licenses flagged as "verify before vendoring." The two highest-confidence near-term wins are the timestamp-supervision label format (low effort, fixes the boundary gap) and Two-Stage AL for label selection; B-ACT and the self-training methods are higher-payoff research bets.